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SQL Course

(Basic to Advance)

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Agenda at a glance

1. Installing MySql and Creating a DB
2. Inserting Data
3. DQL & DML Operations
4. Aggregate Functions & Subqueries
5. Joins & Grouping
6. Window Functions & Stored Procedures

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Must Do

Complete this Chapter at one stretch

Strictly, should not get up while learning

If you feel tired, stop this and start tomorrow.

But, complete this at one stretch.

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Must Have

1.
2.
3.

4.
5.

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Before We begin...

"Small steps each day lead to big results tomorrow."

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Chapter 1

Installing MySQL and Creating DB

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Agenda at a glance

Chapter 1

1.1 Installing MySQL

1.2 Creating a DB

1.3 Theory

Less than 1 Hour

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SQL

Structured Query Language

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Data

What is Data ?

- Data refers to facts, information, or observations
- Data can take various forms and may include numbers, text, images, tables etc

Types of data

- **Structured :** Structured data is organized into well-defined formats, such as databases with rows and columns
- **Unstructured Data :** Unstructured data lacks a specific structure and can include things like text documents, emails, and multimedia content

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Data Base

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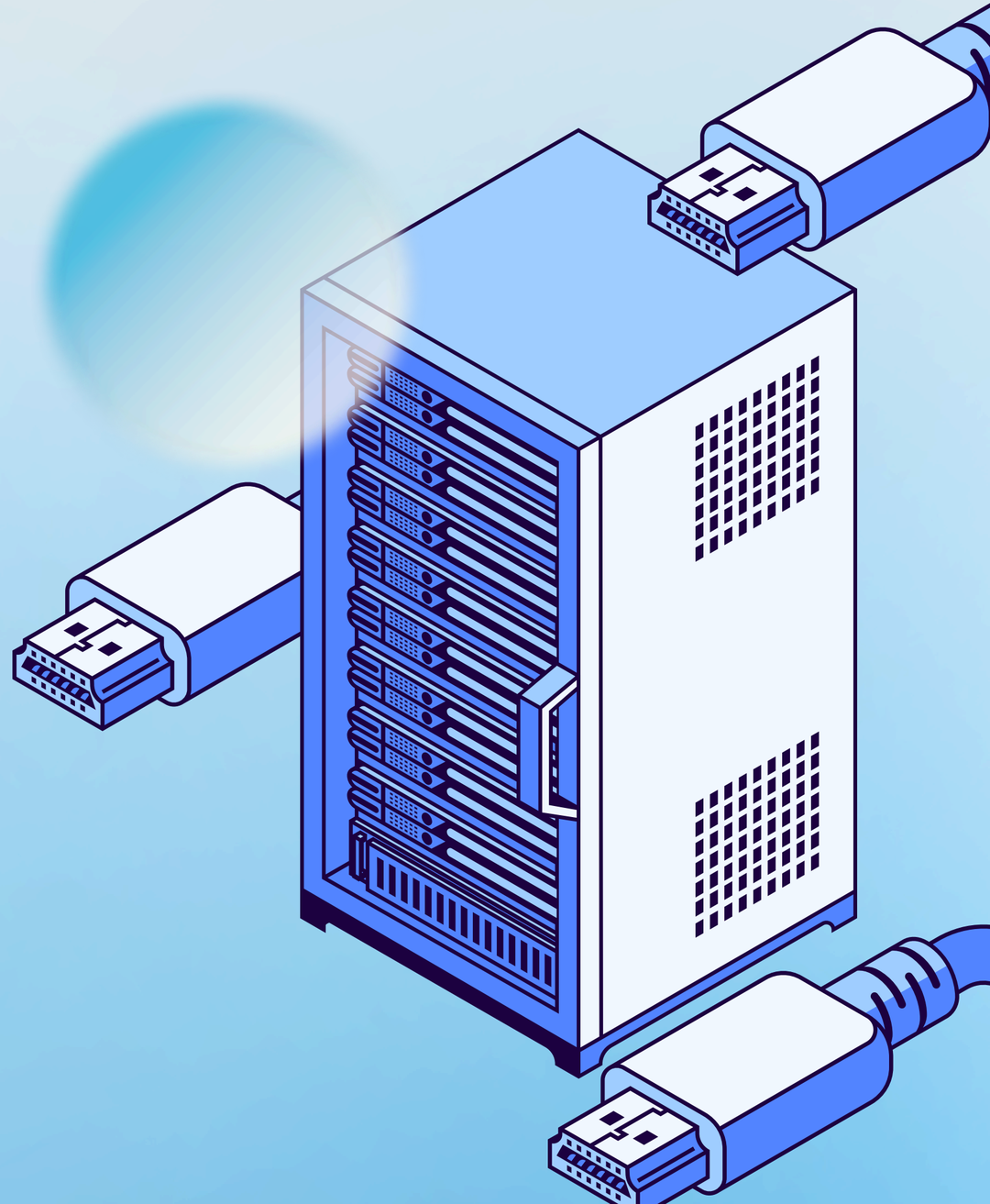
Where do we store Data?

Excel- 10,00,000 rows, and the column 16,384.

SQL



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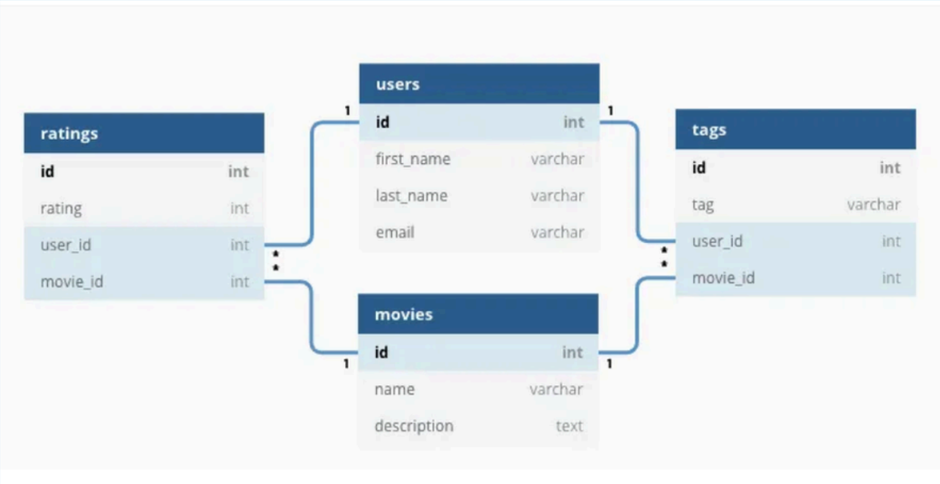
SQL(Structured Query Language)



SQL is a language used to interact and manage Relational databases

Tools used to manage Relational Data base is known as "**Relational Database Management System**"

MySQL
PostgreSQL
Oracle Database
Microsoft SQL Serve

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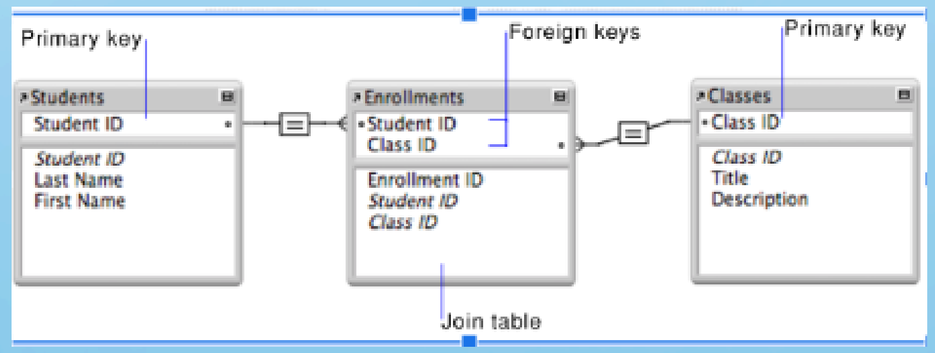
Import Terminologies

Table: It stores data in rows and columns

ID	NAME	CLASS	MARK	GENDER
1	John Deo	Four	75	female
2	Max Ruin	Three	85	male
3	Arnold	Three	55	male
4	Krish Star	Four	60	female
5	John Mike	Four	60	female
6	Alex John	Four	55	male
7	My John Rob	Five	78	male
8	Asruid	Five	85	male
9	Tes Qry	Six	78	male
10	Big John	Four	55	female

Primary Key: A unique identifier for each row in a table, ensuring that each row is distinct.

Foreign Key: A column or set of columns in one table that refers to the primary key in another table, establishing a relationship between the two tables



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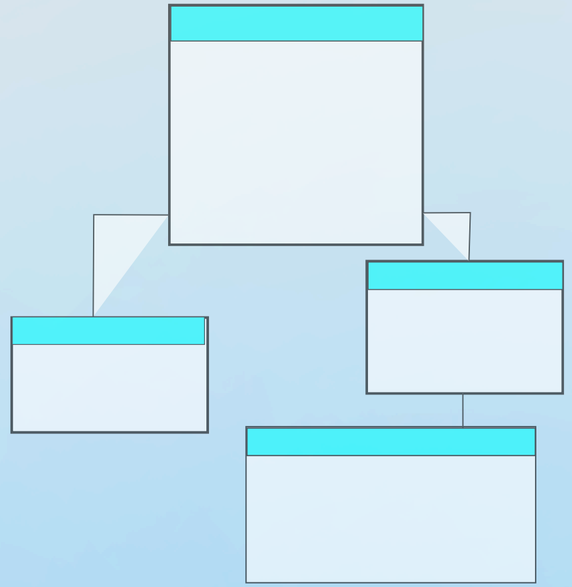
students

studentid	firstname	lastname	BirthDate	gender
1	John	Doe	2000-05-15	Male
2	Jane	Smith	1999-04-12	Female
3	Emily	Johnson	2001-07-22	Female
4	Michael	Williams	2000-12-30	Male
5	Sarah	Brown	1998-10-10	Female
6	David	Jones	2002-03-25	Male
7	Emma	Garcia	2000-11-08	Female
8	James	Martinez	1999-01-01	Male
9	Olivia	Hernandez	2001-08-30	Female
10	William	Lopez	2000-02-14	Male
NULL	NULL	NULL	NULL	NULL

courses

CourseID	CourseName	Credits
1	Mathematics	3
2	Computer Science	4
3	Biology	3
4	Chemistry	4
5	Physics	3
6	Literature	2
7	History	3
8	Economics	3
9	Engineering	4
10	Data Science	4
NULL	NULL	NULL



enrollment

EnrollmentID	StudentID	CourseID	EnrollmentDate
1	1	1	2021-08-20
2	1	2	2021-08-20
3	2	1	2021-08-20
4	2	3	2021-08-20
5	3	4	2021-08-20
6	4	2	2022-01-15
7	5	3	2021-08-20
8	6	5	2022-01-15
9	7	6	2021-08-20
10	8	7	2022-01-15
11	9	8	2021-08-20

Enrollment 7

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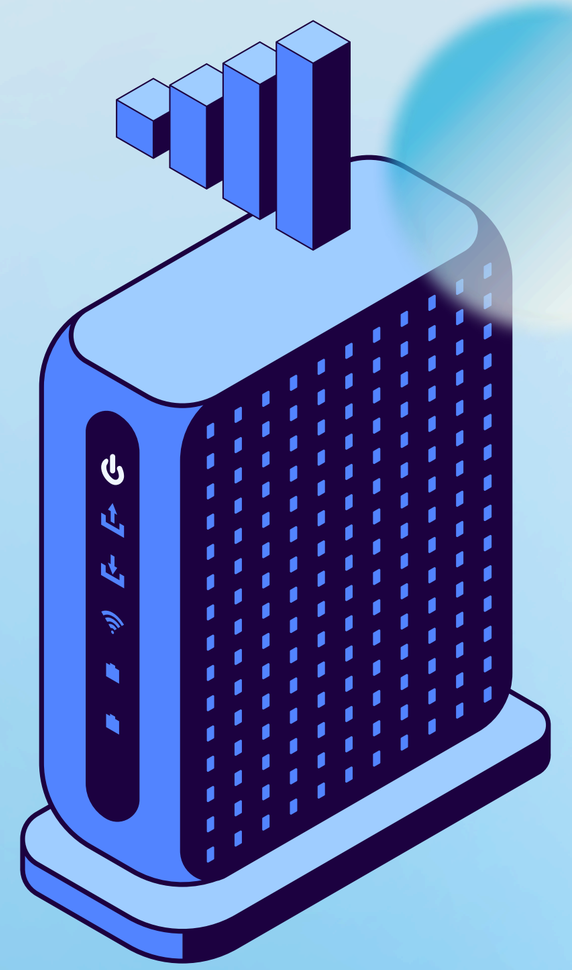
SQL Data Types

- **Numeric Data Types**
  - **Integer**
    - Int eg- 123
    - BigInt eg- 123466789456
  - **Floating**
    - Float eg- 3.15
    - Double eg- 3.145835
  - **Fixed Point**
    - Decimal (10,2) eg- 12345678.90
- **String Data Types**
  - **Character**
    - Char(n) eg- ‘OpenAi’
    - VarChar(n) ‘Hello World’
  - **Text**
    - Text ‘Hii this is an example for the text’
- **Date/Time Data Types**
  - Date eg 2024-11-01
  - Date Time eg 2024-11-01 12:30:45
  - Timestamp eg 2024-11-01 12:30:45
- **Boolean Data Type**
  - True
  - False
- **JSON/XML Data Types**
  - Json {Key:Value}
  - XML (e.g., <note><to>User</to><body>Hello</body></note>)
- **Money Data Type**
  - Money $ 200



BY

Nasir
NASIR

CTO ProitBridge
Data Scientist

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Keyword	Purpose
CREATE	Create a new database, table, or other database object.
ALTER	Modify the structure of an existing table or database object.
DROP	Permanently delete a database, table, or database object.
TRUNCATE	Remove all data from a table without deleting its structure.
RENAME	Rename a table or database.
SELECT	Retrieve data from a table or database.
INSERT	Add new records to a table.
UPDATE	Modify existing records in a table.
DELETE	Remove specific records from a table.
GRANT	Assign privileges to a user or role.
REVOKE	Remove privileges from a user or role.

Keyword	Purpose
START TRANSACTION	Begin a database transaction.
COMMIT	Save changes made during a transaction.
ROLLBACK	Undo changes made during a transaction.
SAVEPOINT	Set a point within a transaction to roll back to.
SET TRANSACTION	Define properties (e.g., isolation level) for a transaction.
PRIMARY KEY	Define a unique identifier for each row in a table.
FOREIGN KEY	Enforce relationships between tables.
UNIQUE	Ensure all values in a column are unique.
NOT NULL	Prevent null values in a column.
CHECK	Validate data based on a condition.
DEFAULT	Assign a default value to a column if no value is specified.
INNER JOIN	Retrieve matching rows from multiple tables.

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Keyword	Purpose
LEFT JOIN	Retrieve all rows from the left table and matching rows from the right.
RIGHT JOIN	Retrieve all rows from the right table and matching rows from the left.
FULL JOIN	Retrieve all matching and non-matching rows from both tables.
WHERE	Filter rows based on conditions.
GROUP BY	Group rows based on a column or expression.
HAVING	Filter groups after grouping them with GROUP BY.
ORDER BY	Sort rows in ascending or descending order.
LIMIT	Restrict the number of rows returned in a query.
OFFSET	Skip a specified number of rows before returning results.
DISTINCT	Select unique records from a column or table.
CASE	Implement conditional logic in queries.
IFNULL	Return a default value if a column is NULL.

Keyword	Purpose
COALESCE	Return the first non-NULL value from a list.
EXPLAIN	Analyze and debug query performance.
SHOW	Display database metadata (e.g., tables, indexes).
DESCRIBE	Display the structure of a table.
CREATE INDEX	Create an index to improve query performance.
DROP INDEX	Remove an index from a table.
USE	Select a database to work with.

Summary

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Chapter 1

1.1 Installing MySQL

1.2 Creating a DB

1.3 Theory

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Chapter 2

Inserting and selecting Data

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Agenda at a glance

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Chapter 2

2.1 Inserting Data

2.2 Selecting Data

2.3 Theory

Less than 1 Hour

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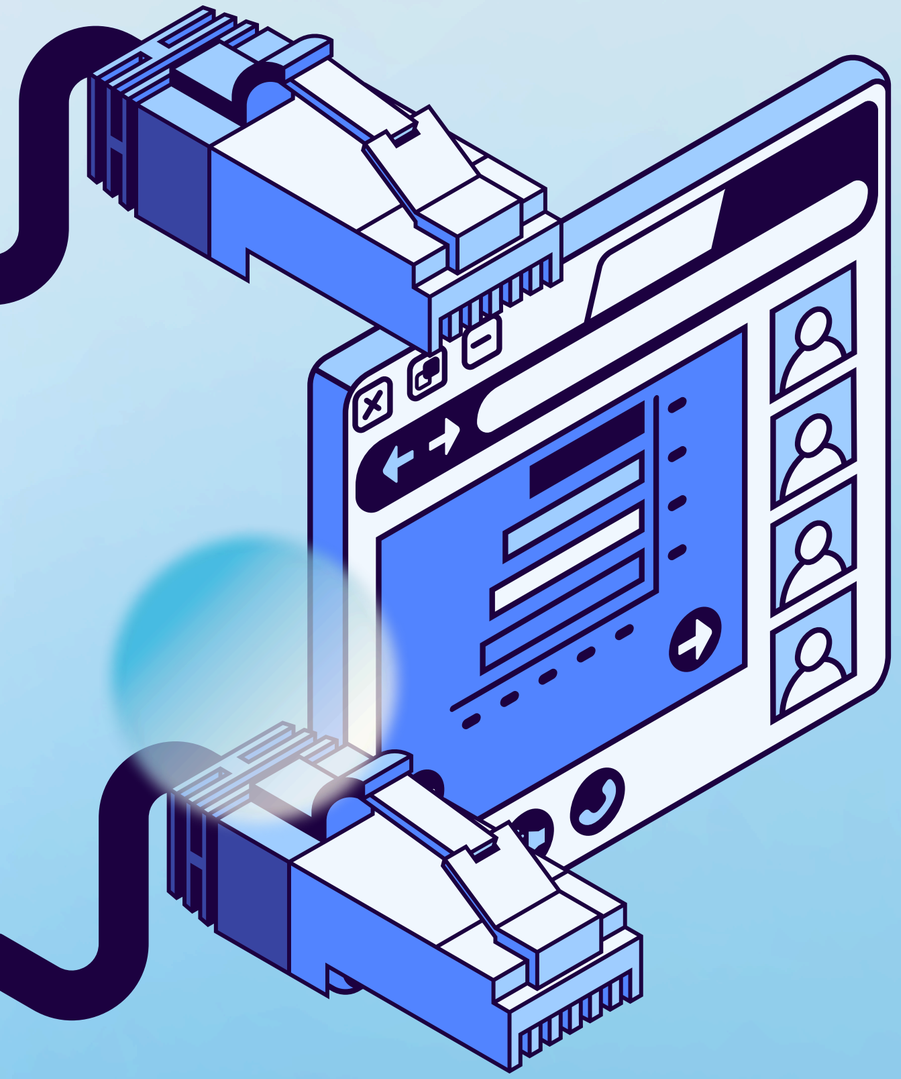

In SQL, there are four main categories of SQL statements, each serving a different purpose:

Data Definition Language (DDL)

Data Manipulation Language (DML)

Data Control Language (DCL)

Data Query Language (DQL)

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Data Definition Language (DDL):

DDL statements are used to define the structure of a database, including creating, modifying, and deleting database objects like tables, indexes, and constraints.

Key DDL statements include:

CREATE: Used to create new database objects, such as tables, indexes, and views.

ALTER: Used to modify the structure of existing database objects.

```sql
CREATE TABLE Students (
    StudentID INT AUTO_INCREMENT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    BirthDate DATE,
    Gender VARCHAR(10)
);
```

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Data Manipulation Language (DML)

DML statements are used to manipulate the data within a database

INSERT: Adds new records to a table.

UPDATE: Modifies existing records in a table.

```sql
INSERT INTO Students (FirstName, LastName, BirthDate, Gender) VALUES
('John', 'Doe', '2000-05-15', 'Male'),
('Jane', 'Smith', '1999-04-12', 'Female'),
('Emily', 'Johnson', '2001-07-22', 'Female'),
('Michael', 'Williams', '2000-12-30', 'Male'),
('Sarah', 'Brown', '1998-10-10', 'Female'),
('David', 'Jones', '2002-03-25', 'Male'),
('Emma', 'Garcia', '2000-11-08', 'Female'),
('James', 'Martinez', '1999-01-01', 'Male'),
('Olivia', 'Hernandez', '2001-08-30', 'Female'),
('William', 'Lopez', '2000-02-14', 'Male');
```

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Data Control Language (DCL):

DCL statements are used to manage permissions and access control within a database.

GRANT: Provides specific privileges to users or roles.

REVOKE: Removes specific privileges from users or roles

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Data Query Language (DQL)

DQL statements are used to query and retrieve data from the database.

SELECT: Retrieves data from one or more tables based on specified criteria.

```sql
SELECT * FROM Students;

SELECT * FROM Courses;

SELECT * FROM Enrollment;

SELECT FirstName, LastName FROM Students;

SELECT * FROM Students WHERE Gender = 'Male';
```

Sequence of execution

Step	Clause	Purpose
1	**FROM**	Specifies the data source (tables, views, joins). Resolves the tables and retrieves initial data.
2	**WHERE**	Filters rows based on the specified condition.
3	**GROUP BY**	Groups rows into subsets based on one or more columns for aggregation.
4	**HAVING**	Filters groups created by GROUP BY based on a condition.
5	**SELECT**	Selects specific columns, calculated expressions, or aggregated values for the output.
6	**ORDER BY**	Sorts the result set based on specified columns in ascending (ASC) or descending (DESC) order.
7	**LIMIT**	Restricts the number of rows returned in the result set.

Miscellaneous

Alter Vs Update

Aspect	ALTER	UPDATE
Purpose	Modify the structure of the table.	Modify the data in the table.
Scope	Affects columns, constraints, or table schema.	Affects rows (the actual data).
Usage	DDL (Data Definition Language).	DML (Data Manipulation Language).
Effect on Data	Does not change the data.	Changes the actual data values.
Examples	Add/drop/rename/modify columns.	Update specific or all row values.

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Summary

Chapter 2

2.1 Inserting Data

2.2 Selecting Data

2.3 Theory

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Chapter 3

DQL & DML Operations

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Agenda at a glance

Chapter 3

3.1 DQL and DML Operations

Less than 1 Hour

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Data Manipulation Language (DML)

DML statements are used to manipulate the data within a database

INSERT: Adds new records to a table.

UPDATE: Modifies existing records in a table.

```sql
INSERT INTO Students (FirstName, LastName, BirthDate, Gender) VALUES
('John', 'Doe', '2000-05-15', 'Male'),
('Jane', 'Smith', '1999-04-12', 'Female'),
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('Michael', 'Williams', '2000-12-30', 'Male'),
('Sarah', 'Brown', '1998-10-10', 'Female'),
('David', 'Jones', '2002-03-25', 'Male'),
('Emma', 'Garcia', '2000-11-08', 'Female'),
('James', 'Martinez', '1999-01-01', 'Male'),
('Olivia', 'Hernandez', '2001-08-30', 'Female'),
('William', 'Lopez', '2000-02-14', 'Male');
```

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Data Query Language (DQL)

DQL statements are used to query and retrieve data from the database.

SELECT: Retrieves data from one or more tables based on specified criteria.

```sql
SELECT * FROM Students;

SELECT * FROM Courses;

SELECT * FROM Enrollment;

SELECT FirstName, LastName FROM Students;

SELECT * FROM Students WHERE Gender = 'Male';
```

Summary

Chapter 3

3.1 DQL and DML Operations

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Chapter 4

Aggregate Functions & Subqueries

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Agenda at a glance

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Chapter 4

4.1 Aggregate Functions

4.2 Aggregate Functions with WHERE

4.3 Aggregate Functions with GroupBy

4.4 Subqueries

Less than 1 Hour

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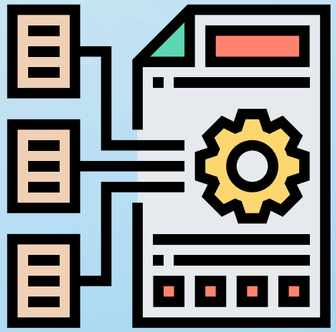
Aggregate functions



Function	Description
COUNT()	Returns the total number of rows.
SUM()	Returns the sum of numeric values.
AVG()	Returns the average of numeric values.
MIN()	Returns the smallest value in a column.
MAX()	Returns the largest value in a column.
GROUP_CONCAT()	Concatenates values into a single string, separated by a comma.
VARIANCE()	Returns the population variance of numeric values.
VAR_POP()	Returns the population variance (equivalent to VARIANCE()).
VAR_SAMP()	Returns the sample variance of numeric values.
STD()	Returns the population standard deviation (alias for STDDEV_POP()).
STDDEV()	Returns the population standard deviation (alias for STDDEV_POP()).
STDDEV_POP()	Returns the population standard deviation.
STDDEV_SAMP()	Returns the sample standard deviation.

Sequence of execution

Step	Clause	Purpose
1	**FROM**	Specifies the data source (tables, views, joins). Resolves the tables and retrieves initial data.
2	**WHERE**	Filters rows based on the specified condition.
3	**GROUP BY**	Groups rows into subsets based on one or more columns for aggregation.
4	**HAVING**	Filters groups created by GROUP BY based on a condition.
5	**SELECT**	Selects specific columns, calculated expressions, or aggregated values for the output.
6	**ORDER BY**	Sorts the result set based on specified columns in ascending (ASC) or descending (DESC) order.
7	**LIMIT**	Restricts the number of rows returned in the result set.

Summary

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Chapter 4

4.1 Aggregate Functions

4.2 Aggregate Functions with WHERE

4.3 Aggregate Functions with GroupBy

4.4 Subqueries

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Chapter 5

Joins & Grouping

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Agenda at a glance

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Chapter 5

5.1 Types of Joins

Less than 1 Hour

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Join

Join Type	Description	Returns
INNER JOIN	Returns only the matching rows from both tables.	Give me what is common in both tables.
LEFT JOIN	Returns all rows from the left table and matched rows from the right table.	All rows from the left table and matched rows from the right.
RIGHT JOIN	Returns all rows from the right table and matched rows from the left table.	All rows from the right table and matched rows from the left.
FULL JOIN	Returns all rows when there is a match in either the left or right table.	All rows from both tables, with NULLs where no match exists.
CROSS JOIN	Returns the Cartesian product of both tables (all combinations of rows).	Every combination of rows from both tables.
SELF JOIN	Joins a table with itself, often used for hierarchical relationships.	Rows from the same table compared to each other.

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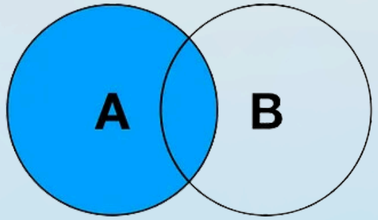
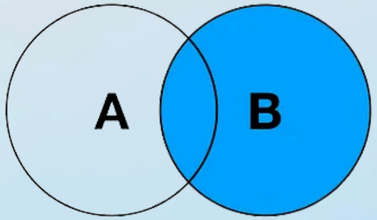
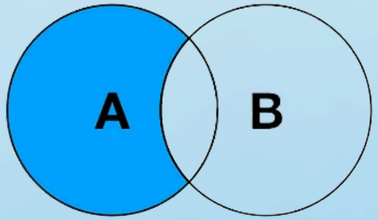
SQL JOINS



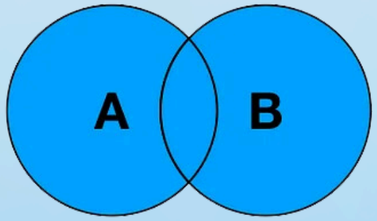
LEFT JOIN



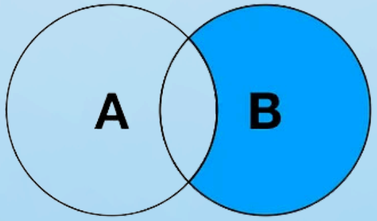
RIGHT JOIN



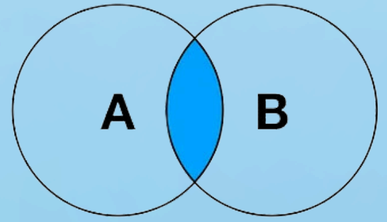
LEFT JOIN EXCLUDING
INNER JOIN



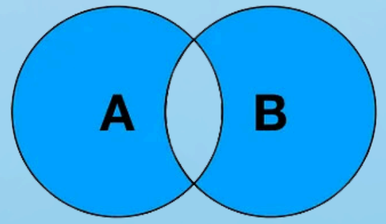
FULL OUTER JOIN



RIGHT JOIN EXCLUDING
INNER JOIN



INNER JOIN



FULL OUTER JOIN EXCLUDING
INNER JOIN

maicmi

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Students

Left

StudentID	Name	CourseID
1	Alice	101
2	Bob	102
3	Charlie	NULL

Courses

Right

CourseID	CourseName
101	Mathematics
102	Science
103	History

Join Type	Explanation	Result Example
INNER JOIN	Returns only students who are enrolled in a course (match in both tables).	Output: Alice (Mathematics), Bob (Science)
LEFT JOIN	Returns all students, even if they're not enrolled in a course. Missing courses will show as NULL.	Output: Alice (Mathematics), Bob (Science), Charlie (NULL)
RIGHT JOIN	Returns all courses, even if no student is enrolled. Students without courses will show as NULL.	Output: Alice (Mathematics), Bob (Science), NULL (History)
FULL JOIN	Combines results of LEFT and RIGHT joins. Shows all students and all courses, with NULL for unmatched entries.	Output: Alice (Mathematics), Bob (Science), Charlie (NULL), NULL (History)
CROSS JOIN	Returns all possible combinations of students and courses.	Output: Alice (Math), Alice (Science), Alice (History), Bob (Math), Bob (Science), Bob (History), etc.
SELF JOIN	Compares rows within the same table (e.g., students mentoring each other).	Example: Find students who share the same CourseID (e.g., Alice mentors Bob if they both have CourseID 101).

Summary

Chapter 5

5.1 Types of Joins

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Chapter 6

Window Functions & Stored Procedures

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Agenda at a glance

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Chapter 6

6.1 Window Functions

6.2 Stored Procedures

Less than 1 Hour

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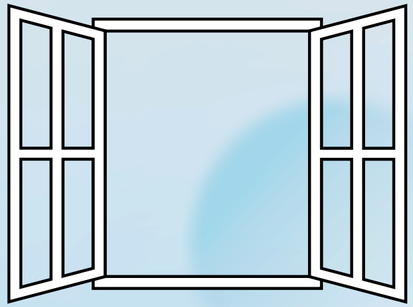

Window Functions

Window functions in SQL are a powerful tool that allow you to perform calculations across a set of table rows related to the current row without collapsing the result set. They are called "window" functions because they operate on a specific range or window of rows, defined by the OVER() clause.

Without Window Functions

department	total_salary
HR	12000
IT	17000
Sales	13000

Window Functions

employee_id	name	department	salary	total_salary_by_department
1	Alice	HR	5000	12000
2	Bob	HR	7000	12000
3	Charlie	IT	8000	17000
4	David	IT	9000	17000
5	Eva	Sales	6000	13000
6	Frank	Sales	7000	13000

Summary

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Chapter 6

6.1 Window Functions

6.2 Stored Procedures

Summary

1. Installing MySql and Creating a DB
2. Inserting Data
3. DQL & DML Operations
4. Aggregate Functions & Subqueries
5. Joins & Grouping
6. Window Functions & Stored Procedures

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